

HoloLearn

User Research Report — Empathize Phase

35 Survey Responses · February – March 2026

35

Total responses

57%

STEM/Eng students

3.8

Avg hologram appeal /5

91%

Would recommend

This report documents the Empathize phase of the HoloLearn design process. It covers the user research conducted via Google Form survey, synthesizes findings into an Empathy Map, defines three User Personas, and presents the key insights and design implications that justify the holographic AI learning concept.

1. User Research

Survey methodology & findings

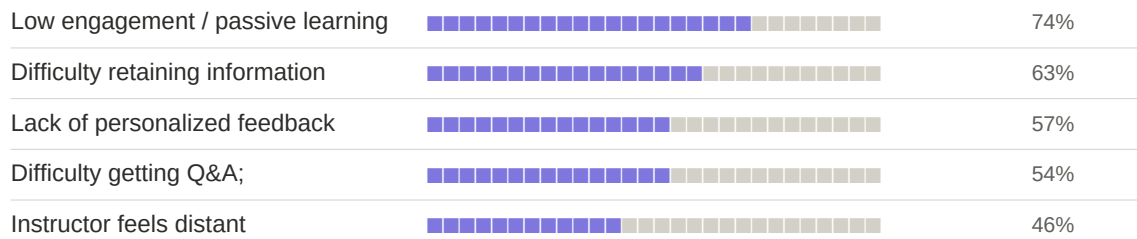
Methodology

User research was conducted via a structured Google Form survey distributed to students, educators, and working professionals. The survey covered current learning habits, platform satisfaction, challenges, attitudes toward AI and holographic technology, and feature priorities. A total of **35 responses** were collected between February 28 and March 5, 2026.

Participant Profile

The sample is dominated by university students aged 19–25 (83%), with a strong STEM/Engineering presence (57%). 91% of respondents identify as female. Most use hybrid learning environments (71%) and access digital learning daily (49%) or 3–4 times per week (31%).

Top Challenges in Current Digital Learning



THINKS & FEELS

"Making digital education more immersive with a Q&A; chatbot that replies instantly."

- Frustrated that passive video lectures don't lead to real understanding
- Anxious about being left behind when instructor is unreachable
- Excited by AI tools, but skeptical of hallucinations and reliability
- Wants to feel personally supported as a learner
- Hopes technology can replicate the energy of a real classroom

SEES

- Hybrid/online learning widely adopted — 71% in hybrid environments
- Classmates passively watching lecture videos without engaging
- AI tools (ChatGPT, tutors) gaining adoption all around them
- Low Q&A; quality on platforms (avg satisfaction 1.8/5)
- Technology gaps — some peers have great resources, others don't

HEARS

"Increase interactivity — we need more than slides and a chatbox."

- Instructors say 'check the recordings' instead of engaging live
- Peers complaining about poor Q&A; during online sessions
- Industry buzz about immersive learning that still feels far away
- Advice to 'be more self-disciplined' but tools don't support it
- Growing academic/tech discussions about VR/AR learning

SAYS & DOES

- Uses YouTube, PDFs, and AI chatbots alongside official platforms
- Accesses learning daily (49%) or 3-4 times per week (31%)
- Mixes AI tools to fill gaps the LMS leaves open
- Shares course materials informally with peers outside the platform
- Attends live or recorded sessions depending on schedule

PAINS

- No one to ask when stuck — timely Q&A; rated 4.6/5 importance
- Content feels generic; no adaptation to learning pace or style
- Instructor feels distant — avg satisfaction 1.4/5
- Cost and limited availability could block access to holographic tools
- Risk of technology being more distracting than helpful

GAINS

- Immersive, memorable learning experiences (top-ranked benefit)
- AI that answers instantly and is grounded in course material
- Sense that an instructor is present and genuinely responsive
- Variety of materials: videos, concept maps, quizzes, guides
- Personalized pacing and feedback — on their own schedule

3. User Personas

Derived from response clusters

Three personas were identified from the survey data, representing the primary user archetypes for the HoloLearn platform.

Persona 1: The Driven STEM Student

MK

Maya Khalil

University Student · STEM / Engineering · Age 21

Experience	3–5 years
Environment	Hybrid (in-person + online)
Frequency	Daily / 3–4 times per week
Tools used	YouTube, AI chatbots, PDFs, LMS
Hologram appeal	4.5 / 5
Preferred setting	Classroom or computer lab

"I want digital education to be more immersive chatbot that replies instantly and doesn't hallucinate variety of materials to study from."

Goals

- Understand complex engineering concepts quickly and deeply
- Get immediate, reliable answers when stuck — no waiting for office hours
- Feel like a real instructor is present, not just a screen
- Access diverse formats: videos, concept maps, interactive guides

Frustrations

- Passive video lectures don't build real comprehension
- Q&A; on platforms is slow or absent — satisfaction drops
- Instructor feels disconnected; no sense of presence
- AI tools are helpful but risk giving wrong course answers

High interest

Immersive learning

Cost concern

Instant Q&A; critical

Persona 2: The Curious Non-STEM Learner



Nour Rashad

University Student · Non-STEM (Humanities / Business) · Age 22

Experience	1–3 years
Environment	Hybrid / fully in-person
Frequency	3–4 times per week
Tools used	LMS, YouTube, AI chatbots, PDFs
Hologram appeal	3.2 / 5
Preferred setting	Home or classroom

"I want more personalized feedback and some to about my problems, and interactive sessions schedule — especially during exams."

Goals

- Receive feedback that feels personal, not generic
- Access learning on a flexible schedule that fits
- Engage with content in novel, exciting ways —
- Have someone (or something) to talk through concepts with

Frustrations

- Lack of personalized feedback makes it hard to
- Low engagement with current platforms — feels through motions
- Traditional in-person sessions don't adapt to he
- Prefers human interaction; skeptical of fully aut

Moderate interest	Engagement focus	Cost barrier	Traditional preference
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Persona 3: The Tech-Forward Educator

SA

Sara Abdulaziz

Educator / Corporate Trainer · Age 29

Experience	1–3 years
Environment	Hybrid
Frequency	Daily
Tools used	Webinars, AI chatbots, LMS, PDFs
Hologram appeal	5 / 5
Preferred setting	Community center / corporate facility

"Making learning materials — even books — more interactive. Students need to feel immersed and not just passive receivers of information."

Goals

- Make course content engaging and memorable
- Reduce cognitive distance between instructor and remote/hybrid students
- Leverage AI to create adaptive, high-quality materials
- Improve learner outcomes and reduce the need for tutoring

Frustrations

- Current platforms create a one-way broadcast, not dialogue
- Hard to monitor engagement or know who is truly learning
- Lacks tools to quickly convert course content into different formats
- Cost and lack of integration with existing systems are major blockers

Very high interest

Presence-focused

Adoption champion

Integration concern

4. Research Report

Rationale & design implications

Why This Idea Was Selected

Digital learning platforms are widely adopted but deeply unsatisfying. Across all 35 respondents — primarily university students aged 19–25 in hybrid environments — a clear pattern emerged: existing tools create passive, disconnected experiences. **74% cited low engagement** as their top challenge, **63% struggle to retain information**, and instructor satisfaction averaged a poor **1.4 out of 5**. The core problem is not the quantity of digital content, but the **absence of presence, interaction, and personalization**.

Key Findings

- 1 Engagement is broken.** 74% of users experience passive, low-engagement learning. Current LMS platforms receive poor interactivity ratings (avg 1.3/5). Students heavily supplement with YouTube and AI chatbots — signaling the gap left by official tools.
- 2 Q&A; is critical and broken.** 91% rated the ability to ask questions during learning as 'important' or 'very important' (4–5/5). Yet Q&A; satisfaction on current platforms averaged just 1.8/5 — the lowest-rated feature by far.
- 3 Instructor presence matters.** 46% cited 'instructor feels distant' as a key pain. Respondents strongly value visual/animated explanations and instructor presence, indicating they learn better when they feel someone is there with them.
- 4 Holographic tech has high appeal but one major barrier.** 77% said immersive learning was their top expected benefit, with an average hologram appeal rating of 3.8/5. However, 77% flagged cost as their primary concern — making accessibility central to product strategy.
- 5 AI content generation is broadly accepted.** 71% preferred AI-generated lectures from uploaded materials. 89% want explanations with examples and 83% want answers grounded in course materials — a strong validation of the RAG-powered Q&A; concept.

Why Holographic AI Is the Right Direction

The survey validates three pillars of the HoloLearn concept. First, **presence**: users don't just want information — they want to feel taught. A holographic AI tutor directly addresses the 'instructor feels distant' pain raised by 46% of respondents. Second, **instant, reliable Q&A;**: the desire for immediate, course-grounded answers (4.6/5 importance) is precisely what an AI Q&A; system can deliver at scale. Third, **immersion**: the top-ranked benefit users expect is a more immersive and memorable learning experience — the core promise of holographic presentation.

Design Implications

Holographic experience

Prioritize **clarity and content pacing** (rated 4.2/5 importance) above raw visual spectacle. Users want to learn, not just be entertained. Include animated explanations as a core delivery mode. The system must work across home, classroom, and computer lab settings.

AI Q&A; system

Must be **immediate** (54% want answers within seconds, 26% within 1 minute) and **grounded in course content** to avoid hallucinations — a specific, repeated concern. Include conversation history and follow-up question suggestions as differentiating features.

Content creation

AI-generated lectures from uploaded materials are preferred by 71% — validating the core workflow. Offer rich material variety: structured PDFs, visual flowcharts/concept maps, concise summaries, quizzes, worksheets, and interactive study guides.

Risks and Mitigation

Cost barrier (77%): Consider tiered pricing, institutional licensing, or university partnerships to reduce individual cost burden.

Limited availability (63%): Cloud-based delivery without requiring specialized hardware is critical for early adoption.

Distraction risk (37%): Design focused learning flows — not an open-ended sandbox. User testing must include attention and comprehension metrics.